

VIREN

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PROFESSIONAL SUMMARY

Results-oriented Technical Leader and Solution Architect with multi-year experience driving large-scale enterprise data architecture, low-latency streaming analytics, and AI-first initiatives across complex distributed sectors. Proven expert in building secure, resilient data pipelines that transform raw transaction streams and high-volume telemetry events into highly available indexing solutions and real-time reporting APIs. Exceptionally skilled in test-driven infrastructure orchestration, advanced data quality observability, and close cross-functional collaboration with backend engineering teams to design reliable data schemas.

PROFESSIONAL EXPERIENCE

UnitedHealth Group | United States

Sept 2016 – Present

Lead Architect

- **On-Chain & IoT Data Pipelines:** Design, build, and operate robust, end-to-end low-latency data pipelines turning raw chain events, smart contract logs, and device telemetry streams into production-grade analytics datasets.
- **Custom Indexers & Subgraphs:** Develop and maintain highly optimized subgraphs (The Graph, Ponder) and custom RPC indexers to extract, parse, and map multi-chain datasets for client-facing dashboards and real-time APIs.
- **Data Quality, Observability & Alerting:** Own comprehensive data validation, pipeline observability, and custom alerting frameworks across critical production stacks to guarantee high availability and absolute transactional integrity.
- **Telemetry & Edge-to-Cloud Integration:** Integrate complex sensor and device telemetry data streams with blockchain-anchored state architectures, enabling seamless real-world tracking and reliable streaming analytics (Kafka, Redis, Postgres).
- **Schema & Contract Design Collaboration:** Partner directly with senior blockchain engineers, backend developers, and dApp specialists to design interoperable data schemas, optimize database models, and review contract telemetry points.
- **Infrastructure & Test-Driven Mindset:** Orchestrate secure, scalable microservices architectures using Docker and major cloud platforms (AWS/GCP) following a strict engineering-first, test-driven approach.

Exelon Corp | PA, MD

Mar 2008 – Aug 2016

Architect

- **Process Evaluation & System Architecture:** Evaluated legacy data workflows and directed systemic architecture modernizations utilizing JIRA and HP QC tracking platforms to systematically boost pipeline reliability.
- **Enterprise Integration Layers:** Architected the corporate REST API tech stack to operate as a high-throughput proxy gateway, securely routing application data packages between operational teams and primary data clusters.
- **Security & Policy Management:** Configured application integrations with Layer 7 policy managers, enforcing compliance protocols using secure authorization frameworks (OAuth1.0/2.0), LDAP directories, and corporate PKI data encryption structures.

Chicago Mercantile Exchange (CME) | IL, USA

Nov 2006 – Feb 2008

Sr. Consultant

- **High-Volume Processing Environments:** Streamlined low-latency stream processing configurations by optimizing the FIX Fast Protocol Converter, processing millions of complex transaction records for high-volume market reporting datasets.
- **Schema Design & Automation:** Designed TIBCO BusinessWorks workflows and complex XML schemas to establish clean Service-Oriented Layers utilizing Java and Spring frameworks to safely automate massive enterprise data movements.
- **System Performance Diagnostics:** Maintained continuous system health monitoring and provided tier-3 production troubleshooting for high-availability database environments to prevent service dropouts and data discrepancies.

Independence Community Bank / Sovereign Bank | NY

Mar 2004 – Oct 2006

Sr. Consultant

- **Multi-Store Synchronizations:** Developed a comprehensive CI Integrity system to synchronize and validate customer records across distributed legacy databases, maintaining strict data quality standards.
- **Middleware Messaging Infrastructures:** Architected middleware request routing layers using JMS BusinessWorks and Tibco, ensuring transactional state durability and smooth data orchestration across decentralized applications.

British Petroleum (BP) | UK

Feb 2003 – Jun 2003

Sr. Developer

- **Corporate Data Integration:** Built and supported distributed corporate data integration layers (IL), designing robust rule bases to manage pipeline component health and tracking logs.

Hutchison3G | UK

Mar 2003 – May 2003

Sr. Analyst Programmer

- **Message Bus Implementations:** Implemented data access solutions via a message bus, optimizing process flow codebases and configuring repositories for high-availability load balancing and fault tolerance.

Convergys Information Management Group | FL

Dec 2000 – Feb 2003

Sr. C++/Java Programmer

- **Raw Stream Extraction:** Built high-performance C++ sockets and dynamic libraries to ingest, parse, and process millions of raw network activity logs and usage receipts for high-volume billing indexers.

Auric Software Ltd. | India

May 1999 – Nov 2000

Software Engineer

- **Relational Code Optimization:** Designed relational internet software packages backed by optimized Oracle schemas, utilizing rigorous engineering patterns to maximize complex computational script speeds.

EDUCATION

BS in Computer Engineering — Pune University

Diploma in Advanced Computing — CDAC, Pune